Fiscal Year:	FY 2022	**Task Last Updated:**	FY 03/31/2022
PI Name:	Laurie, Steven Ph.D.		
Project Title:	Distribution of Body Fluids during Long Duration Space Flight and Subsequent Effects on Intraocular Pressure and Vision Disturbance		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	281-244-0029
Organization Name:	KBR/NASA Johnson Space Center		
PI Address 1:	Cardiovascular and Vision Laboratory		
PI Address 2:	2400 NASA Pkwy		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058-2749	**Congressional District:**	36
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2011 Crew Health NNJ11ZSA002NA
Start Date:	08/01/2020	**End Date:**	03/31/2022
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:	May 2022 update per PI: Several of the CoInvestigators on this project have overlapped with "sister projects" (other Fluid Shifts investigations) under PI: Hargens and PI: Dulchavsky. Three CoInvestigators left the project during this reporting period: Smith Johnston, Ph.D., Christine Ribeiro, Ph.D., and Babs Soller, Ph.D. (Ed., 5/20/22)		
COI Name (Institution):	Johnston, Smith M.D. (NASA Johnson Space Center) Lee, Stuart Ph.D. (KBR/NASA Johnson Space Center) Martin, David M.S. (KBR/NASA Johnson Space Center) Ploutz-Snyder, Robert Ph.D. (Universities Space Research Association, Columbia) Smith, Scott Ph.D. (NASA Johnson Space Center) Soller, Babs Ph.D. (Reflectance Medical Inc.) Marshall-Goebel, Karina Ph.D. (KBR/NASA Johnson Space Center) Ribeiro, Laura Christine J.D., M.S. (KBR/NASA Johnson Space Center) Stenger, Michael (NASA Johnson Space Center) Arbeille, Phillippe (University of Tours) Dulchavsky, Scott (Henry Ford Health System) Ebert, Douglas (KBR/NASA Johnson Space Center) Hargens, Alan (University of California, San Diego)		

	Macias, Brandon (NASA Johnson Space Center) Zwart, Sara (NASA Johnson Space Center) Sargsyan, Ashot (KBR/NASA Johnson Space Center)
Grant/Contract No.:	Internal Project
Performance Goal No.:	
Performance Goal Text:	
Task Description:	NOTE: Continuation of "Distribution of Body Fluids during Long Duration Space Flight and Subsequent Effects on Intraocular Pressure and Vision Disturbance" with new Principal Investigator (PI) Dr. Steven Laurie. Previous PI was Dr. Michael Stenger, who became Human Health & Countermeasures Element Scientist. See Cumulative Bibliography under Stenger for additonal publications. The central objective of the proposed work is to characterize the magnitude of the headward fluid shift during long duration space flight and to measure the subsequent compartmentalization of this fluid. The second objective is to correlate in-flight alterations of eye structure, ocular vascular parameters, and vision with headward fluid shifts, vascular dimensions, and flow patterns. Finally, the third objective is to determine systemic and ocular factors of individual susceptibility to the development of these inflight alterations. In order to determine the effect of space flight on the headward fluid shift and subsequent redistribution across fluid compartments, biochemical, ultrasound, and other non-invasive cardiovascular measures will be performed before launch, during flight, and after landing. Total body water, extracellular, and intracellular fluid volume will be determined by biochemical dilution techniques. Ultrasound will be used to assess upper vs. lower body interstitial fluid and vascular flow and structure changes, ocular dimensions and optic nerve sheath diameter, and jugular venous pressure. Near infrared spectroscopy will be used to measure lower vs. upper body local tissue hydration, and total peripheral resistance will be calculated from noninvasively acquired blood pressure and cardiac output. It is currently unclear why only some astronauts experience vision disturbances with space flight. It is hypothesized that astronauts respond differently to the fluid shift inflight and that this impacts changes in ocular structure and function changes. The goal of this study is to test this hypothesis. The research proposed here will determine the effect of long duration space flight on fluid shifts and subsequent redistribution across fluid compartments and attempt to discern the contribution of these adaptations to changes in ocular structure and function. NOTE: This study was merged with investigations from Dr. Alan Hargens (Fluid distribution before, during and after prolonged space flight) and Dr. Scott Dulchavsky (Microgravity associated compartmental equilibration) resulting in a comprehensive study titled “Fluid Shifts Before, During and After Prolonged Space Flight and Their Association with Intracranial Pressure and Visual Impairment” (short title: Fluid Shifts).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	An early hypothesis considered elevated intracranial pressure (ICP) as a cause for space flight associated neuro-ocular syndrome. Because of this, novel noninvasive ICP techniques included tympanic membrane displacement and otoacoustic emissions are being investigated to determine ICP changes associated with space flight. Knowledge gained from this study may inform medical professionals treating and studying patients suffering from idiopathic intracranial hypertension, a debilitating condition with some characteristics in common with that experienced by astronauts.
Task Progress:	May 2022 Addendum An update was made to the final bibliography of the associated project under Michael Stenger, Ph.D. Please see "Distribution of Body Fluids during Long Duration Space Flight and Subsequent Effects on Intraocular Pressure and Vision Disturbance" (PI: Stenger). Below is an update from the “Fluid Shifts” technical progress report added as per PI: Laurie (Ed., 5/20/22). The Fluid Shifts study has greatly enhanced our understanding of numerous aspects of spaceflight physiology and our understanding of Spaceflight Associated Neuro-ocular Syndrome (SANS). • The 12 crewmembers who participated in 52 sessions of lower body negative pressure (25 mmHg) for up to 60 minutes per session tolerated the sessions well without needing to terminate a test. • During spaceflight numerous key outcome measures were similar to values in the supine posture on Earth, and use of lower body negative pressure (LBNP) during spaceflight partially reversed these values, but none reached values measured in the seated upright posture on Earth. • The effects of LBNP appear to be transmitted to the level of the eye. • Our team identified altered cerebral venous blood flow draining the head and, for the first time, observed a venous thrombus in the left internal jugular vein. • We have advanced our understanding of the effects of spaceflight on ocular morphology based on enhanced analyses of optical coherence tomography (OCT) images. • We corroborated previous findings from brain magnetic resonance imaging (MRI) analyses and demonstrated that there is not a strong association between the change in lateral ventricular volume and magnitude of optic disc edema that develops during spaceflight. Within this reporting period the final data set was submitted to the NASA Life Sciences Data Archive (LSDA), 2 manuscripts were submitted for publication in peer-reviewed journals, and 1 abstract was presented at the NASA Human Research Program Investigators’ Workshop. Additionally, abstracts were presented at the 2022 Experimental Biology annual meeting and The Association for Research in Vision and Ophthalmology (ARVO) annual meeting.
Bibliography Type:	Description: (Last Updated: 12/10/2025)

Abstracts for Journals and Proceedings	Greenwald SH, Pardon LP, Ferguson CR, Young M, Laurie SS, Macias BR "Small and shallow optic cup morphology is associated with optic disc edema development during spaceflight. " The Association for Research in Vision and Ophthalmology Annual Meeting, Denver, CO, May 1-4, 2022. Abstracts. The Association for Research in Vision and Ophthalmology Annual Meeting, Denver, CO, May 1-4, 2022. , May-2022
Abstracts for Journals and Proceedings	Lytle JR, Macias BR, Lee SMC, Martin D, Ebert DJ, Hargens AR, Dulchavsky SA, Alferova IV, Stenger MB, Laurie SS. "Cardiovascular responses to mild lower body negative pressure during spaceflight." 2022 Experimental Biology Annual Meeting, Philadelphia, PA, April 2-5, 2022. Abstracts. 2022 Experimental Biology Annual Meeting, Philadelphia, PA, April 2-5, 2022. , Apr-2022
Abstracts for Journals and Proceedings	Laurie SS, Arbeille P, Danielson RW, Ebert DJ, Greenwald S, Jasien J, Kemp D, Lee SMC, Liu J, Lytle J, Macias BR, Marshall-Goebel K, Martin DS, Minkoff L, Pardon LP, Ploutz-Snyder R, Sargsyan A, Hargens AR, Dulchavsky SA, Stenger MB. "Fluid Shifts." 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022. Abstracts. 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022. , Feb-2022
Articles in Peer-reviewed Journals	Pardon LP, Greenwald SH, Ferguson CR, Patel NB, Young M, Laurie SS, Macias BR. "Identification of factors associated with the development of optic disc edema during spaceflight." JAMA Ophthalmol. 2022 Oct 27;140(12):1193-1200. http://dx.doi.org/10.1001/jamaophthalmol.2022.4396 ; PMID: 36301519; PMCID: PMC9614681 , Oct-2022
Articles in Peer-reviewed Journals	Sibony PA, Laurie SS, Ferguson CR, Pardon LP, Young M, Rohlf FJ, Macias BR. "Ocular deformations in spaceflight-associated neuro-ocular syndrome and idiopathic intracranial hypertension." Invest Ophthalmol Vis Sci. 2023 Mar;64(3):32. https://doi.org/10.1167/iovs.64.3.32 ; PMID: 36988950; PMCID: PMC10064934 , Mar-2023
Articles in Peer-reviewed Journals	Ferguson CR, Pardon LP, Laurie SS, Young MH, Gibson CR, Brunstetter TJ, Tarver WJ, Mason SS, Sibony PA, Macias BR. "Incidence and progression of chorioretinal folds during long-duration spaceflight." JAMA Ophthalmol. 2023 Feb 1;141(2):168-75. http://dx.doi.org/10.1001/jamaophthalmol.2022.5681 ; PMID: 36602790 PMCID: PMC9857718 , Feb-2023
Articles in Peer-reviewed Journals	Huang AS, Jalili J, Walker E, Weinreb RN, Laurie SS, Macias BR, Christopher M. "Artificial intelligence deep learning models to predict spaceflight associated neuro-ocular syndrome (SANS)." Am J Ophthalmol. 2025 Oct;278:115-123. https://doi.org/10.1016/j.ajo.2025.06.009 ; PubMed PMID: 40505849; PubMed Central PMCID: PMC12481839 , Oct-2025
Articles in Peer-reviewed Journals	He Y, Karanjia R, Zhang X, Wanderer D, Walker E, Lee SH, Poczatek MJ, O'Grady CS, Marion KM, Young M, Moestl S, Mulder E, Stern C, Sadda SR, Laurie SS, Macias BR, Huang AS. "Optic nerve vasculature and countermeasure assessment in a bedrest analogue of spaceflight-associated neuro-ocular syndrome." Am J Ophthalmol. 2025 Oct;278:317-327. https://doi.org/10.1016/j.ajo.2025.06.032 ; PubMed PMID: 40545016; PubMed Central PMCID: PMC12462775 , Oct-2025
Articles in Peer-reviewed Journals	Svoronos AA, O'Grady CS, Walker E, Afshari NA, Macias BR, Laurie SS, Weinreb RN, Huang AS. "Analysis of spaceflight-associated biometric and refractive changes in astronauts." Am J Ophthalmol. 2025 Aug;276:146-56. http://dx.doi.org/10.1016/j.ajo.2025.04.001 ; PMID: 40194644 , Aug-2025
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